

Md. Aslam Mia

M.Sc. Student in Civil Engineering (Transportation)
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RESEARCH INTERESTS

- Intelligent Transportation Systems (ITS)
- Application of AI & ML in Transportation Engineering
- Connected & Autonomous Vehicles
- Transportation Planning

EDUCATION

<i>Master of Science in Civil Engineering (Transportation)</i> Bangladesh University of Engineering and Technology (BUET)	2026 - Present
<i>Bachelor of Science in Civil Engineering (B.Sc. Engg.)</i> (Major - Transportation Engineering, Minor - Geotechnical Engineering) Bangladesh University of Engineering and Technology (BUET)	Awarded: March 2026 Overall GPA: 3.54/4.0
<i>Higher Secondary School Certificate (HSC)</i> Shahid Syed Nazrul Islam College, Mymensingh	Awarded: April 2020 Overall GPA: 5.0/5.0

RESEARCH EXPERIENCE

- ***Undergraduate Thesis*** (Apr. 2025 – Jun. 2026)
Title: “**Performance Evaluation of Traffic Efficiency and Safety, and RL-Based Adaptive Traffic Signal Control under Different Connected and Autonomous Vehicle Penetration Levels**”
Supervisor: Prof. Dr. Md. Mizanur Rahman
Department of Civil Engineering, Bangladesh University of Engineering and Technology
 - Developed traffic scenarios on both a synthetic Grid 4×4 network and a real-world Dublin City centre (DCC) road network in the SUMO microsimulation environment, tested under varying traffic patterns and environmental conditions (peak hours, roadblocks, special events, adverse weather) in different penetration levels of connected and autonomous vehicles (HDV, mixed, and CAV conditions).
 - Evaluated traffic efficiency using Travel Time Rate (TTR) and traffic safety using Post-Encroachment Time (PET) based conflict analysis across different CAV penetration levels.
 - Trained and evaluated an Independent Deep Q-Network (IDQN) adaptive signal controller against a conventional FixedTime baseline, assessing robustness via a recovery-time metric, the duration required to return to optimal performance after an unseen traffic event; achieved average travel-time reductions of up to 34.3%.

PUBLICATIONS

Journal Manuscripts

1. (Under Review) “Evaluation of RL-Based Adaptive Traffic Signal Control under Different Connected and Autonomous Vehicle Penetration Levels in Mixed Traffic” – submitted to International Conference on Civil Engineering (ICCE) 2026, Dhaka, Bangladesh.

PROFESSIONAL EXPERIENCE

- **Junior Engineer** (Aug. 2026 – Present)
Bureau of Research, Testing and Consultation (BRTC), BUET
Project: “Charges of Vetting BoQ for Civil Works under ESPSN Project DPDC”
 - Review and vet Civil Bills of Quantities (BoQs), including quantity estimation and unit-rate analysis for substation building projects.
 - Verify quantities and unit rates using construction drawings, PWD Schedule of Rates, and market quotations.
 - Contribute to technical reviews, progress documentation, and coordination of project deliverables.
- **Intern** (Nov. 2025 – Dec. 2025)
Public Works Department (PWD), Dhaka, Bangladesh
 - Rotated through the Design Division, Project Division, and Materials Testing Laboratory on active government construction

PROJECT EXPERIENCE

- **Image Annotation for Computer-Vision-Based Traffic Parameter Analysis** (2024 – 2025)
Departmental Senior Thesis Support, BUET
 - Annotated nearly 2,000 video frames from a live intersection in Roboflow to classify pedestrians by age and gender, producing a high-quality dataset used for further model development.
- **Capstone Project - Slum Relocation: Multistoried Residential Building Complex** (2025 – 2026)
 - Designed a low-cost multistoried residential complex as part of a slum relocation initiative, addressing structural, functional, and affordability constraints.
 - Used ETABS and SAP for structural modelling; produced construction drawings in AutoCAD.
 - Developed project planning documentation and engineering design deliverables as part of a team

SKILLS

Technical Skills

- **Simulation Software:** SUMO
- **Design & Drafting Software:** AutoCAD
- **Structural Analysis Software:** ETABS, SAP2000
- **Programming & Data Analysis:** MATLAB, C++, Python, Machine Learning frameworks
- **Productivity Tools:** Microsoft Word, Excel, PowerPoint

Language Proficiency

- **Bangla:** Native
- **English:** Professional working proficiency

CLUB ACTIVITIES

- **General Member**, ACI (American Concrete Institute) - BUET Student Chapter
- **General Member**, ASCE (American Society of Civil Engineers) - BUET Student Chapter

REFERENCES

Dr. Md. Mizanur Rahman

Professor, Department of Civil Engineering
Bangladesh University of Engineering and Technology
(BUET)

Email: mizanur@ce.buet.ac.bd

Relation: Academic (Undergraduate Thesis Supervisor)

Dr. Sumaiya Afroz

Assistant Professor, Department of Civil
Engineering
Bangladesh University of Engineering and
Technology (BUET)

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Relation: Academic (academic advisor of undergrad. studies)